

SHAUNAK RANE

Associate AI Engineer | Agentic AI | LLM Systems | AI Automation

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PROFESSIONAL SUMMARY

AI Engineer with proven experience architecting autonomous multi-agent systems, structured output generation pipelines via LLM APIs, and production-grade RAG retrieval systems. Strong foundation in software engineering fundamentals, object-oriented design, and containerized REST microservices using FastAPI, PostgreSQL, and Docker. Hands-on background developing predictive time-series models for mission-critical industrial assets, with end-to-end focus on API observability, regression testing, and reproducible system promotion.

TECHNICAL SKILLS

AI & LLM Systems: Agentic AI, Multi-Agent Workflows, RAG, Prompt Engineering, Structured Outputs (JSON Schema), Function & Tool Calling, Vector Embeddings, LLM APIs (Gemini, OpenAI)

Machine Learning & Data: PyTorch, PyTorch Geometric, Graph Neural Networks (GraphSAGE), Scikit-learn, Statsmodels, Time-Series Modeling, Anomaly Detection, NumPy, Pandas

Backend & Languages: Python, C, SQL, TypeScript/JavaScript, FastAPI, Flask, RESTful APIs, WebSockets, OOP, System Architecture

Databases & DevOps: PostgreSQL, Supabase (pgvector), MySQL, Docker, Git, GitHub Actions, Linux, Postman, pytest, CI/CD

Engineering Practices: Causal Evaluation, Statistical Quality Gates, API Observability, Test-Driven Development (TDD), Microservices

WORK EXPERIENCE

Univitt Technologies - AI Engineering Intern

May 2026 - Aug 2026 | Thane, India (Remote)

- Architected end-to-end predictive analytics pipeline for centrifugal compressor Condition-Based Monitoring (CBM) across 3 industrial facilities and 4,000+ daily historian records.
- Reproduced 3,879 historical Degradation Factor Index (DFI) rows to $<1e-10$ absolute error; engineered causal maintenance-event logic with 17/17 regression checks passing.
- Built and containerized REST inference microservices in Docker with Flask, automating anomaly detection alerts and cutting operational diagnostic turnaround.
- Audited 400+ physical and mechanical process variables, evaluated ML vs. statistical forecasting baselines, and instituted automated promotion gates for model deployment.

Univitt Technologies - Software Engineering Intern

Apr 2025 - Jul 2025 | Thane, India (Remote)

- Engineered high-throughput RESTful backend APIs and database schemas with FastAPI and PostgreSQL for an institutional food operations and cost-management platform (Sodexo).
- Developed end-to-end automated workflows for menu generation, Bill-of-Materials (BOM) cost calculations, live inventory tracking, and operator dashboards.
- Integrated a Random Forest predictive attendance model into operational shift scheduling, achieving 88% forecast accuracy across workforce shifts.
- Implemented comprehensive test suites with pytest, executed schema migrations, and collaborated on code reviews using Git and GitHub workflows.

PROJECTS

Project Aegis - Autonomous Multi-Agent Triage System

Python, FastAPI, Agentic AI, RAG, Supabase (pgvector), Docker

- Architected an autonomous emergency response orchestrator with hybrid RAG retrieval and tool-calling agent swarms, enforcing strict JSON Schema validation for deterministic structured outputs.
- Engineered tool-calling agent workflows to autonomously query incident records, evaluate casualty severity, and dispatch prioritized emergency actions with sub-second API latency.
- Implemented semantic vector persistence via Supabase (pgvector), API-key isolation, and structured audit trails for agent-to-agent message passing and real-time incident reporting.

Gridium Protocol - Multi-Agent Microgrid Optimization

Python, PyTorch, FastAPI, DDPG, WebSockets

- Developed a multi-agent reinforcement learning engine utilizing Deep Deterministic Policy Gradients (DDPG) to dynamically optimize power routing across a 15-node simulated grid.
- Coordinated decentralized agent policies managing solar generation, battery storage, and dynamic loads, achieving an 18.4% reduction in peak-hour grid strain.
- Built real-time telemetry streaming pipelines using FastAPI and WebSockets, feeding continuous operational metrics to interactive monitoring dashboards.

TopoFlow - GNN Fluid Dynamics Simulation

Python, PyTorch Geometric, GraphSAGE, FastAPI

- Benchmarked GraphSAGE against Kozeny-Carman physics baselines across 1,231 pore-network samples from 5 geological formations, using pore-size heterogeneity to dictate the modeling regime.
- Reduced MSE loss by 46.2% on Savonnières and 28.4% on Estailades formations while maintaining rigorous physical consistency benchmarks across baseline solvers.
- Engineered pore-network extraction pipelines, GNN model training workflows, and FastAPI inference microservices with PyTorch Geometric.

EDUCATION

Universal AI University - B.Tech in Artificial Intelligence & Machine Learning

Expected May 2028 | Karjat, Maharashtra

- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Machine Learning, Deep Learning, Database Management Systems (DBMS), Operating Systems, Linear Algebra & Probability.